

Failure Propagation Chain

A simple chain showing how upstream conditions turn into measurable issues, user pain, and business/mission impact.

LAYER 0

External Drivers

What broad forces are creating risk before metrics move?

How to read it

Left side: the chain itself. Right side: what each layer asks.
Top to bottom: hidden causes -> visible effects.

- Budget pressure or cost cutting
- Understaffed ops / weak ownership
- Launch pressure or deadline rush

LAYER 1

Structural Conditions

What about the design makes the system fragile?

- Too many services depend on one datastore
- No failover or weak redundancy
- Aggressive retries / weak backpressure

LAYER 2

Triggering Events

What event actually started the chain?

- Config error or bad deploy
- Traffic spike
- Dependency slowdown

LAYER 3

Internal Stress Mechanics

How is the system absorbing the shock?

- Retries multiply request volume
- Queues grow faster than workers drain them
- Workers or connections fall behind

LAYER 4

Telemetry Warning Signs

What quantitative data shows the issue?

- Latency climbs
- Timeouts increase
- Cache hit rate drops

LAYER 5

User-Visible Degradation

What does the customer experience?

- Pages load slowly
- Notifications arrive late
- One region feels worse than another

LAYER 6

Business / Mission Impact

Why does this matter?

- Lost revenue
- SLA violations
- Reputational harm

Use this chain to reason from hidden causes to observable symptoms, then to customer and mission impact.